

1. Abstract
 - a. Summarize the entire poster in roughly 200 - 350 words.
2. Introduction
 - a. The first paragraph is about the different grand challenges as a definition and their negative consequences. Also add one scientific picture to prove your points.
 - b. The second paragraph is on two prior solutions. Mention its full name, its function, its pros/strengths, and its cons/weaknesses.
 - c. The third paragraph is on what modifications can be made to improve the prior solutions. So, mention the opposite of its cons/weaknesses.
 - d. The fourth paragraph is on your idea. Start by mentioning how your solution is better than the prior solutions. Then state your idea briefly. Then state your design requirements. Then state how this solution with its different aspects solves the grand challenges that you mentioned in the first paragraph.
3. Materials
 - a. In a table write all the materials that you used in the prototype only. You will have three columns per item: (Item – Usage – Picture)
4. Methods
 - a. Write all of the steps that you took to make this prototype and mention them in chronological order and don't forget to number each step.
 - b. Mention your test plan steps, so the things you will do to test your prototype.
5. Results
 - a. In this section you present all the data you collected from your prototype during your test plan.
 - b. Present this data using both a graph and data table and do not forget to number them.
6. Analysis
 - a. The first paragraph is a brief summary of the grand challenges that you wrote in the first paragraph in the introduction.
 - b. The second paragraph is on how your project solves the grand challenges that you just mentioned.
 - c. The third paragraph is on whether your prototype's results from the test plant met your design requirements. At the end of this paragraph mention the pros and cons of your prototype.
 - d. Use various laws to calculate different things from your results. And after you find a law that you are going to use. First write that law, then use a table to represent the inputs to the law and the outputs that came from that law. Then from that table present the data using a graph. Then after you do that try to conclude something new.
 - e. Your last paragraph will be on learning transfers. So, what did you learn this semester that helped you in this project, this could be related to the prototype itself or the analysis you just did.
7. Conclusion
 - a. Restate that design requirements your project did or didn't meet.
 - b. Mention where you got the idea from (your inspiration) and also mention how it differs and is better than it.
8. Recommendations
 - a. Write a small introduction to your recommendations.

- b. Write 3 – 5 different things you would do if you were to do this project again.
- 9. Literature cited
 - a. Cite at least five sources in (APA 7) format. These five sources have to be related to your project and they have to have variety. So, you should have at least one website, one scientific research paper, and one book.
- 10. Acknowledgement
 - a. In this you can thank all of the people that helped on your project, and yes this can include teachers.
- 11. For further information
 - a. Put the school emails of your entire team.